

Set 12: Properties and structures of atoms

Set 12: Exercises

- The Atom Timeline: A student searches for images of the structure of the atom and finds many different images, including those on the right. Answer the following questions about the structure of the atom.
 - Why are there different representations of the atom?
 - What do the different images convey about the understanding of the structure of the atom?
 - Why do you think the structure shown on page 128 is more accurate than structure A?
 - Sequence the structures to make a timeline of our understanding of atomic structure. Include the structure on page 128.
 - Briefly explain what evidence was used by scientists to determine the different structural representations shown.
- Find out how emission spectroscopy is used to help identify plastics for recycling.
- Describe how the following isotopes are used in medical research:
 - Technetium-99
 - Molybdenum-99
 - Cobalt-60.
- Each of the following scientists have made significant contributions to our understandings of the atomic model. For each scientist, describe their contribution, including a diagram where appropriate.

Dalton
Thomson
Rutherford
Bohr
Chadwick
- Under what circumstances would atomic absorption spectroscopy (AAS) be used?
- With the aid of a diagram explain how an AAS works.

